Tutorial 8

Indian Institute of Technology Jodhpur
Mathematics-I (MAL1010)
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1. Find the length of the following curves, on the given domain:
   (a) $y = \frac{1}{x}$, $x \in [1, 3]$
   (b) $y = \ln(\cos x)$, $x \in \left[0, \frac{\pi}{4}\right]$
   (c) Cycloid $x = r(t - \sin t),\ y = r(1 - \cos t)$, $t \in [0, 2\pi]$

2. Calculate the area between or enclosed by the following curves/parametric equations:
   (a) $y = x^2$ and $y = 2x$, $x \in [0, 2]$
   (b) $y = e^x$ and $y = x + 2$, $x \in [0, 1]$
   (c) $y = \sqrt{x}$ and $y = x$, $x \in [0, 1]$
   (d) Astroid $x = \cos^3 t,\ y = \sin^3 t,\ t \in [0, 2\pi]$

3. Compute the volume of the solid generated by revolving the following around the indicated axes and domains:
   (a) $y = \sqrt{x}$, about the x-axis, $x \in [0, 4]$
   (b) Aread bounded by $y = 2x - x^2$ and below by the x-axis, over the interval $x \in [0, 2]$
   (c) Ellipse $x = \cos t,\ y = \sin t,\ t \in [0, \pi/4]$, about the y-axis
   (d) Parametric $x = t - \sin t,\ y = 1 - \cos t$, about the x-axis, $t \in [0, 2\pi]$

4. Find the surface area generated by revolving the following about the x-axis (or specified axis):
   (a) $y = \sqrt{x}$, $x \in [0, 1]$
   (b) Astroid $x = a\cos^3 t,\ y = a\sin^3 t,\ t \in [0, \frac{\pi}{2}]$
   (c) $y = 3x^{1/3}$ for $0 \le y \le 3$ $(0 \le x \le 1),$ about the y-axis

5. A water tank in the shape of a right circular cone has height h and base radius r. Compute the surface area (excluding the base) using integration, and find the total area if the base is included.

6. A manufacturer wants to paint a cylindrical pipe of height h and radius r. Use integration to compute the lateral surface area, and then determine the total painted area if both the top and bottom surfaces are also included.

7. The frustum of a cone has lower base radius r_1, upper base radius r_2 $(r_2 < r_1)$, and slant height s. Set up and compute the surface area of the frustum using the integral formula for surfaces of revolution (excluding the bases).